AI Buildathon

Theme Brief: Tool for Students

ANU AIMSOC & ANUEC

Central Challenge

Create a practical tool that helps university students solve a real academic, administrative, organisational, or campus-life problem.

Do not build a generic chatbot. Build a focused student tool with a clear workflow and demonstrable value.

1. Theme Overview

The theme of the AI Buildathon is "Tool for Students". Participants are challenged to design and build a practical tool that addresses a specific problem experienced by university students. The tool may support academic work, administrative navigation, personal organisation, campus engagement, student services, career planning, accessibility, or any other meaningful aspect of university life.

The central expectation is that teams build something useful, specific, and demonstrable. Participants may use AI to develop their solution and may also include AI within the product itself, but the project should not be a generic AI chatbot. A strong project should have a defined student user, a concrete problem, a clear input, a structured workflow, a useful output, and honest limitations.

The four focus areas are designed to guide ideation while still allowing creativity. Each team should choose one focus area and build a solution that directly responds to that student situation.

Your team has just been hired by **ANU**.

The university is placing a stronger focus on improving the everyday education experience of its students. Across campus, students are facing challenges that are often easy to overlook but difficult to ignore once you notice them.

Some students are trying to understand degree rules, course requirements, forms, and policies that feel more complicated than they should. Some are trying to fit lectures, tutorials, assessments, part-time work, commuting, and sleep into the same week - usually with sleep making the biggest sacrifice. Some are working hard but still struggling to understand difficult course content. Others are looking for support, opportunities, study groups, or a sense of community, but do not know where to start.

ANU does not want another long survey, another forgotten feedback form, or another strategy document that lives peacefully in a PDF folder.

They want something built.

A solution that helps with one real student problem.

Something a student could open on a Monday morning when they are trying to plan their week, use it on a Wednesday afternoon when they are stuck on course content, or come back on a Thursday night when they realise they need help before the deadline arrives.

Your challenge is to identify one meaningful student problem and build a tool that makes that problem easier, clearer, or more manageable.

2. Core Principles for All Projects

- **Student-specific problem:** The tool should solve a problem that is clearly connected to university student life. It should not be a generic productivity, finance, or chatbot app with superficial student branding.
- **Clear workflow:** The tool should guide a student from input to outcome. Judges should be able to understand what the student enters, what the system does, and what useful output the student receives.
- **Practical value:** The tool should help the student take action, make a decision, understand something, connect to support, or complete a task more effectively.
- **Responsible use of AI:** AI can be used for explanation, recommendation, summarisation, planning, matching, or feedback, but teams should recognise limitations and avoid misleading or unsafe outputs.
- **Honest limitations:** Teams should clearly state what the tool can and cannot do, especially where official university rules, academic integrity, privacy, or personal support are involved.

3. Focus Areas Summary

Focus Area	Student situation	Expected project direction
The Confused Student	A student is unsure how to navigate rules, courses, policies, forms, or admin processes.	Build a tool that converts confusion into clear next steps.
The Overwhelmed Student	A student is struggling with deadlines, workload, competing tasks, or limited time.	Build a tool that organises, prioritises, and creates a realistic action plan.
The Struggling Learner	A student is having difficulty understanding course content or preparing for assessments.	Build a tool that supports learning, practice, feedback, and conceptual understanding.
The Isolated Student	A student is disconnected from people, services, groups, events, opportunities, or support.	Build a tool that connects the student to relevant resources or communities.

4. Focus Area 1: The Confused Student

Help students understand what to do next.

This subtheme focuses on students who are blocked because university systems are difficult to understand. The problem may involve course selection, degree progression, administrative forms, extension processes, special consideration, enrolment rules, prerequisite requirements, graduation checks,

Example student problems

- Which courses should I take next semester?
- Am I meeting my degree requirements?

Example tool directions

- Course planner
- Degree requirement checker

What a strong project should do

- Ask for relevant student context, such as degree, course code, deadline, completed courses, or issue type.
- Interpret a rule, process, or requirement in plain language.
- Give specific next steps such as documents needed, contact points, options, warnings, or verification steps.
- Avoid pretending to be an official university authority; include a verification reminder where appropriate.

5. Focus Area 2: The Overwhelmed Student

Help students manage workload, deadlines, and priorities.

This subtheme focuses on students who have too much to manage at once. They may be balancing multiple assignments, exams, classes, part-time work, group tasks, personal commitments, or conflicting deadlines. A strong project should turn a messy workload into a realistic plan that helps the student decide what to do first.

Example student problems

- I have multiple assignments due in the same week.
- I work part-time and need a realistic study plan.

Example tool directions

- Weekly study planner
- Timetable optimiser

What a strong project should do

- Collect concrete workload information, such as tasks, due dates, weights, progress, difficulty, and available time.
- Break large tasks into smaller, manageable steps.
- Prioritise using clear logic such as urgency, weight, time required, risk, and dependencies.
- Produce a realistic plan rather than an idealised schedule that ignores constraints.

Help students understand difficult course content.

This subtheme focuses on students who are finding course material difficult. The challenge may involve understanding lectures, formulas, readings, programming concepts, exam topics, or prerequisite knowledge. A strong project should improve learning rather than simply producing final answers.

Example student problems

- I want to practise before the exam.
- I do not know which topics I am weak in.

Example tool directions

- Lecture-to-study-guide generator
- Exam revision coach

What a strong project should do

- Diagnose what the student currently understands or does not understand.
- Explain concepts at an appropriate level using examples, steps, or analogies.
- Include practice questions, feedback, or checks for understanding.
- Encourage active learning by guiding students through the learning process rather than simply providing direct answers.

7. Focus Area 4: The Isolated Student

Help students connect with support and opportunities.

This subtheme focuses on students who feel disconnected from people, services, groups, events, opportunities, or university support systems. The issue may be academic, social, career-related, campus-related, or support-related. A strong project should connect students to something specific and useful, not just present a static list of links.

Example student problems

- I want to find a study group.
- I am new to campus and feel lost.

Example tool directions

- Study group matcher
- Campus services recommender

What a strong project should do

- Ask for relevant context, such as course, interests, availability, location, year level, goals, or support need.
- Recommend specific people, groups, services, events, opportunities, or resources.
- Explain why each recommendation fits the student.
- Give a next step such as join, message, book, attend, apply, or contact.

8. Participant Expectations

Each team should choose one subtheme and design a tool that is specific enough to be meaningful within the 10-hour buildathon. The final prototype does not need to be production-ready, but it should demonstrate a clear student use case and a working or convincingly interactive workflow.

- **Specific student problem:** The team should be able to state the exact student problem in one sentence.
- **Clear input:** The tool should ask for the information needed to solve the problem.
- **Useful workflow:** The tool should guide the student through a structured process instead of simply producing generic advice.
- **Practical output:** The final result should be something the student can use: a plan, checklist, recommendation, diagnosis, feedback report, route, match, or next-step guide.
- **Honest limitations:** The team should explain where the tool may be incomplete, uncertain, unofficial, or dependent on better data.

9. What Participants Should Avoid

- **Generic chatbot:** A chatbot that answers any student question without a defined workflow is too broad.
- **Generic productivity app:** A calendar, notes app, or to-do list must include student-specific logic to be relevant.
- **Overpromising:** Tools should not claim official authority over policies, course rules, health, finances, or academic outcomes.
- **Academic misconduct:** Learning tools should support understanding and feedback, not help students submit dishonest work.
- **Too much scope:** A smaller working prototype is better than a large platform that cannot be demonstrated clearly.

Good Luck!